

# Linear Programming: Problem Set

Classics, modelling tricks, simplex pitfalls and proofs

Covers Chapters 1–3 (modelling, equational form, bases and BFS, the fundamental theorem, simplex).

Difficulty: ★★ medium, ★★★ hard. Full solutions start on page 7.

**Notation.** As in the lectures: a tableau for a basis  $B$  is  $x_i = p_i + \sum_{j \in N} q_{ij}x_j$  ( $i \in B$ ),  $v = v_0 + \sum_{j \in N} r_j x_j$ ;  $r_j$  are the reduced costs; the support of  $x$  is  $S(x) = \{j : x_j > 0\}$ ;  $a_j$  is column  $j$  of  $A$ . Unless stated otherwise, simplex uses the largest positive reduced cost to choose the entering variable.

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## Part I — Modelling tricks

### Problem 1 Absolute values: a warehouse on a highway ★★

Five towns lie along a straight highway at kilometre marks  $a = (0, 20, 30, 70, 100)$ . Each day,  $w = (2, 1, 3, 1, 2)$  trucks go from the warehouse to the respective town. The company wants to place the warehouse at a mark  $x$  minimising the total truck-kilometres  $f(x) = \sum_i w_i |x - a_i|$ .

- Explain why  $f$  is not linear. Introduce variables  $e_i$  with  $e_i \geq x - a_i$  and  $e_i \geq a_i - x$  and write an LP.
- Explain why, at every optimal solution of your LP,  $e_i = |x - a_i|$  for all  $i$ .
- Solve the problem. *Hint:*  $f$  is piecewise linear; compute its slope on each interval.
- A zoning rule forces  $50 \leq x \leq 80$ . What is the new optimum and its cost?
- A colleague wants to place a noisy depot as *far* as possible from the towns and writes  $\max \sum_i w_i e_i$  over the same constraints, with  $0 \leq x \leq 100$ . What goes wrong, and why?

### Problem 2 Min–max: the best line in the worst case ★★★

Fit a line  $y = \alpha + \beta x$  to the four points  $(0, 0), (1, 2), (2, 1), (3, 4)$  so that the *largest* absolute residual  $\max_i |y_i - \alpha - \beta x_i|$  is as small as possible (Chebyshev fit).

- Write this as an LP in the variables  $\alpha, \beta, t$ . Which variables are free in sign? How many variables would the equational form have?
- Denote  $\rho_i = y_i - \alpha - \beta x_i$ . Find nonnegative weights  $\lambda$  on three of the constraints “ $\pm \rho_i \leq t$ ” that eliminate  $\alpha$  and  $\beta$ , and deduce that  $t \geq 1$  for every feasible  $(\alpha, \beta, t)$ .

- (c) Find the optimal line and prove that it is unique.
- (d) Write (without solving) the LP for the least-absolute-deviations fit  $\min \sum_i |\rho_i|$ . Why is least squares,  $\min \sum_i \rho_i^2$ , not an LP?

**Problem 3** Piecewise-linear costs: an electricity tariff

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A workshop makes lamps of type A and B. Net of materials, an A sells for 13 DH and a B for 9 DH. An A needs 2 kWh and 1 labour-hour, a B needs 1 kWh and 1 labour-hour. There are 60 labour-hours this week and the market takes at most 40 lamps of type B. Electricity is billed in blocks: the first 40 kWh at 1 DH/kWh, the next 60 kWh (from 40 to 100) at 3 DH/kWh, and everything beyond 100 kWh at 7 DH/kWh.

- (a) Let  $E$  be the energy used and  $C(E)$  the bill. Why is the profit not a linear function of the production?
- (b) Model the problem as an LP using block variables  $e_1 \in [0, 40]$ ,  $e_2 \in [0, 60]$ ,  $e_3 \geq 0$  with  $e_1 + e_2 + e_3 = E$ .
- (c) Nothing in the LP says “fill block 1 before block 2”. Prove that for fixed production the cheapest admissible split fills the blocks in order, and conclude that the LP solves the real problem.
- (d) Solve. *Hint*: write  $13x_1 + 9x_2$  as a combination of  $E = 2x_1 + x_2$  and  $L = x_1 + x_2$ .
- (e) The utility now offers a *volume discount*: 5 DH/kWh for the first 40 kWh, 2 DH/kWh beyond. Show that the same block model gives a wrong answer, and explain which property of the tariff made (c) work.

**Problem 4** Ratios: the Charnes–Cooper transformation

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A lab chooses  $x_1, x_2 \geq 0$  (hundreds of hours on two methods) subject to

$$x_1 + 3x_2 \leq 9, \quad 2x_1 + x_2 \leq 8,$$

and wants to maximise the efficiency ratio

$$R(x) = \frac{2x_1 + 3x_2 + 1}{x_1 + x_2 + 2}.$$

- (a) Why is this not an LP?
- (b) Substitute  $t = \frac{1}{x_1 + x_2 + 2}$  and  $y = tx$ . Show that this leads to the LP
- $$\max 2y_1 + 3y_2 + t \quad \text{s.t.} \quad y_1 + 3y_2 \leq 9t, \quad 2y_1 + y_2 \leq 8t, \quad y_1 + y_2 + 2t = 1, \quad y, t \geq 0.$$
- (c) Prove that the two problems are equivalent: show that every feasible  $(y, t)$  has  $t > 0$ , and give mutually inverse maps between the feasible sets that preserve the objective.
- (d) Solve, and prove optimality with a one-line inequality.
- (e) Where did you use that the denominator is positive on the feasible set?

**Problem 5** Percentages: blending tea

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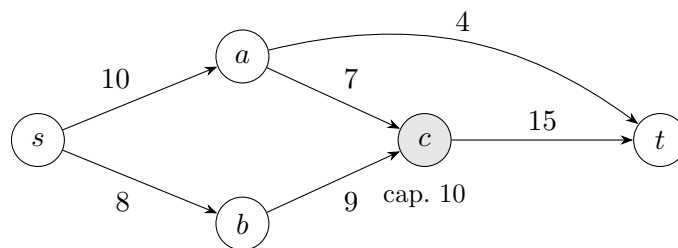
A shop blends green tea (cost 60 DH/kg, 30 kg in stock), black tea (40 DH/kg, 50 kg) and mint (25 DH/kg, 80 kg), and sells any amount of the blend at 55 DH/kg. The recipe requires that green tea makes up *at least 25%* of the blend by weight and mint *at most 40%*. The total amount produced is a decision.

- Write the LP. Turn the percentage rules into linear constraints and discuss the case of an empty blend.
- Show, directly from the constraints, that no feasible plan produces more than 120 kg.
- Find the optimal blend and prove optimality with nonnegative multipliers on the constraints.
- Green tea loses 5 DH on every kilogram sold. Why does the optimal plan use all of it? Estimate how much profit one extra kilogram of green tea in stock would bring.

**Problem 6** Networks: node capacities in a maximum flow

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Data flows from  $s$  to  $t$  in the network below (arc labels are capacities in MB/s). The router  $c$  can process at most 10 MB/s in total, whatever the arcs allow.



- Write the maximum-flow LP, including the router constraint.
- Replace  $c$  by two nodes  $c_{\text{in}} \rightarrow c_{\text{out}}$  joined by an arc of capacity 10. Show that the flows correspond, and explain what this buys you compared with (a).
- Find a maximum flow and prove it is maximum with a cut.
- The operator may add one unit of capacity to exactly one arc (or to the router). Which choices raise the maximum flow? Justify every answer.

**Part II — Geometry, bases and the simplex method****Problem 7** Redundant rows, singular pairs and a degenerate vertex

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Consider  $\max 5x_1 + 2x_2 + x_3 + x_4 + 3x_5$  over  $x \in \mathbb{R}^5$ ,  $x \geq 0$ , with

$$\begin{aligned} x_1 + x_2 + x_3 + 2x_5 &= 4 \\ 2x_1 + x_2 + x_4 + 2x_5 &= 8 \\ 3x_1 + 2x_2 + x_3 + x_4 + 4x_5 &= 12. \end{aligned}$$

- Row-reduce  $[A \mid b]$ , find  $\text{rank } A$  and decide what to do with the redundant equation. What would happen if the third right-hand side were 13?
- After the reduction there are  $\binom{5}{2} = 10$  candidate column pairs. For each, decide whether it is a basis, compute the basic solution, and decide feasibility.

- (c) The 10 candidates give only 4 distinct BFS. Account for the difference using the three distinct failure modes.
- (d) Show that  $x^* = (4, 0, 0, 0, 0)$  is optimal. Write the objective row for each of the four bases that represent  $x^*$ . Which of them prove optimality? Is  $x^*$  the unique optimum?
- (e) Start the simplex method from the basis  $\{x_1, x_2\}$ . What happens on the first pivot?

**Problem 8** Alternative optima and a sensitivity range

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max  $2x_1 + 4x_2$  s.t.  $x_1 + 2x_2 \leq 8$ ,  $x_1 \leq 4$ ,  $x_2 \leq 3$ ,  $x \geq 0$ .

- (a) Run the simplex method from the slack basis until the optimality test holds.
- (b) The final tableau has a zero reduced cost. Pivot on it. What changes and what does not?
- (c) Describe the set of all optimal solutions and prove your description from the tableau.
- (d) Replace the objective by  $c_1x_1 + 4x_2$ . For which  $c_1$  is  $(2, 3)$  optimal? Which vertex is the unique optimum for  $c_1$  slightly above 2?

**Problem 9** An unbounded region and a parametric objective

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Let  $X = \{x \in \mathbb{R}^2 : x \geq 0, -x_1 + x_2 \leq 2, x_1 - 2x_2 \leq 2\}$  and consider max  $x_1 + \alpha x_2$  over  $X$ , where  $\alpha \in \mathbb{R}$  is a parameter.

- (a) Sketch  $X$ , list its vertices, and show that  $X$  is unbounded.
- (b) A *recession direction* is a  $d \neq 0$  with  $x + \theta d \in X$  for all  $x \in X$ ,  $\theta \geq 0$ . Show that the recession directions are exactly  $d = \lambda(1, 1) + \mu(2, 1)$ ,  $\lambda, \mu \geq 0$ , not both 0.
- (c) Write the tableau of the basis  $\{x_1, s_1\}$  (slacks  $s_1, s_2$ ) for general  $\alpha$  and use it to show: the LP is unbounded if and only if  $\alpha > -2$ . Give an explicit improving ray.
- (d) For  $\alpha < -2$  and for  $\alpha = -2$ , find *all* optimal solutions.
- (e) For  $\alpha = -2$  the optimal set is unbounded but the optimal value is finite. Does this contradict the fundamental theorem? Which part of Theorem 3.3.9 is at work here?

**Problem 10** No starting basis: the two-phase method

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min  $3x_1 + 2x_2$  s.t.  $x_1 + x_2 \geq 4$ ,  $x_1 + 3x_2 \geq 6$ ,  $x_1 \leq 5$ ,  $x \geq 0$ .

- (a) Put the problem in equational form with surpluses  $s_1, s_2$  and a slack  $s_3$ . Show that the basis  $\{s_1, s_2, s_3\}$  is infeasible.
- (b) *Phase I*. Add artificial variables  $a_1, a_2 \geq 0$  to the two  $\geq$  rows and solve max  $w = -a_1 - a_2$ . Explain why this auxiliary LP has an obvious feasible basis, why its optimum is  $\leq 0$ , and why the original LP is feasible if and only if the optimum is 0.
- (c) Run Phase I (drop an artificial as soon as it leaves the basis).
- (d) *Phase II*. Continue from the basis found, with the true objective. Give the optimum and a certificate.
- (e) Replace  $x_1 \leq 5$  by  $x_1 + x_2 \leq 3$  and run Phase I again. Conclude.

## Part III — Proofs

### Problem 11 When are the linearisation tricks correct? ★★

Let  $X \subseteq \mathbb{R}^n$  and let  $f_1, \dots, f_k$  be affine functions.

- Prove that  $\inf_{x \in X} \sum_i |f_i(x)| = \inf \left\{ \sum_i e_i : x \in X, e_i \geq f_i(x), e_i \geq -f_i(x) \right\}$ , and that  $(x, e)$  is optimal for the right-hand problem iff  $x$  is optimal for the left and  $e_i = |f_i(x)|$ .
- State and prove the analogous result for  $\min_{x \in X} \max_i f_i(x)$  and for  $\max_{x \in X} \min_i f_i(x)$ .
- Take  $n = 1$ ,  $X = [-1, 1]$  and the problem  $\max_{x \in X} |x|$ . Show that the construction of (a) with “max” gives an unbounded LP. Then prove the stronger statement: there is *no* LP in variables  $(x, z)$  whose set of optimal solutions projects onto the set of maximisers  $\{-1, 1\}$ . (You may use Problem 14.)

### Problem 12 Weak duality and complementary slackness ★★★

Let  $(P)$ :  $\max c^\top x$  s.t.  $Ax \leq b$ ,  $x \geq 0$ , and call  $y$  a *certificate* if  $y \geq 0$  and  $A^\top y \geq c$ .

- Prove: for every feasible  $x$  and every certificate  $y$ ,  $c^\top x \leq b^\top y$ .
- Prove: if  $c^\top x = b^\top y$  then  $x$  is optimal for  $(P)$  and  $y$  minimises  $b^\top y$  over certificates.
- Prove complementary slackness: if  $c^\top x = b^\top y$  then  $y_i (b - Ax)_i = 0$  for all  $i$  and  $x_j (A^\top y - c)_j = 0$  for all  $j$ .
- Apply this to
 
$$\max 4x_1 + 3x_2 + 5x_3 \text{ s.t. } x_1 + x_2 + 2x_3 \leq 10, \quad 2x_1 + x_2 + x_3 \leq 12, \quad x_1 + 3x_3 \leq 9, \quad x \geq 0$$
 with  $y = (2, 1, 0)$ : find the optimal value and *all* optimal solutions.
- Prove: if  $(P)$  is unbounded, no certificate exists.

### Problem 13 Structure of basic feasible solutions ★★★

Let  $A \in \mathbb{R}^{m \times n}$  have rank  $m$  and  $P = \{x : Ax = b, x \geq 0\}$ .

- Prove that every BFS is an extreme point of  $P$ : it is not  $\lambda u + (1 - \lambda)w$  with  $u \neq w$  in  $P$  and  $0 < \lambda < 1$ .
- Let  $x \in P$  have linearly independent support columns. Prove that  $x$  is the only point of  $P$  whose support is contained in  $S(x)$ . Deduce that two BFS with the same support are equal.
- Deduce that  $P$  has at most  $\binom{n}{m}$  vertices.
- Prove that if every BFS is nondegenerate, then distinct feasible bases give distinct BFS. Show by an example from Part II that this fails without nondegeneracy.

### Problem 14 Convexity of the optimal set ★★

Let  $X = \{x \in \mathbb{R}^n : Ax \leq b\}$  and  $X^* = \{x \in X : c^\top x = \max_{z \in X} c^\top z\}$  (assume the maximum is attained).

- Prove that  $X^*$  is convex.

- (b) Deduce that an LP has no optimal solution, exactly one, or infinitely many.
- (c) Prove that a local maximum of an LP is global: if  $\bar{x} \in X$  and  $c^\top \bar{x} \geq c^\top x$  for all  $x \in X$  with  $\|x - \bar{x}\| < \varepsilon$ , then  $\bar{x} \in X^*$ .
- (d) Show that (c) fails for  $\max x^2$  over  $[-1, 2]$  and for an integer program. Which hypothesis breaks?

**Problem 15** Reduced costs, uniqueness and degeneracy

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Consider a feasible tableau with all reduced costs  $r_j \leq 0$  (so the BFS  $\hat{x}$  is optimal).

- (a) Assume  $\hat{x}$  is nondegenerate ( $p_i > 0$  for all  $i \in B$ ). Prove that  $\hat{x}$  is the unique optimum if and only if  $r_j < 0$  for all  $j \in N$ .
- (b) Show that nondegeneracy cannot be dropped: for  $\max x_1$  s.t.  $x_1 \leq 1$ ,  $x_1 + x_2 \leq 1$ ,  $x \geq 0$ , exhibit an optimal tableau with a zero reduced cost although the optimum is unique.
- (c) Prove: if  $r_k = 0$  and  $q_{ik} \geq 0$  for every  $i \in B$ , then the set of optimal solutions is unbounded.
- (d) Combine with Problem 7(d): what can and cannot be concluded from the signs of the reduced costs at a degenerate optimal vertex?

## Solutions

### Solution to Problem 1

(a)  $|x - a_i|$  has a kink at  $a_i$ : its slope is  $-1$  to the left and  $+1$  to the right, so  $f$  is not of the form  $cx + d$ . Since  $|u| = \max(u, -u)$ , set  $e_i \geq x - a_i$ ,  $e_i \geq a_i - x$ :

$$\min \sum_i w_i e_i \quad \text{s.t.} \quad e_i - x \geq -a_i, \quad e_i + x \geq a_i \quad (i = 1, \dots, 5), \quad x \in \mathbb{R}, \quad e \geq 0$$

( $x$  is free in principle; here  $x \in [0, 100]$  is harmless).

(b) The two constraints say  $e_i \geq |x - a_i|$ . If some  $e_i > |x - a_i|$ , lower it to  $|x - a_i|$ : still feasible, and the objective drops by  $w_i(e_i - |x - a_i|) > 0$  (all  $w_i > 0$ ). So an optimal solution has  $e_i = |x - a_i|$ , and the LP value equals  $f(x)$ . (Problem 11 does this in general.)

(c) On an interval between consecutive towns, the slope of  $f$  is (weight to the left)  $-$  (weight to the right), total weight 9:

| interval | $x < 0$ | $(0, 20)$ | $(20, 30)$ | $(30, 70)$ | $(70, 100)$ | $x > 100$ |
|----------|---------|-----------|------------|------------|-------------|-----------|
| slope    | $-9$    | $-5$      | $-3$       | $+3$       | $+5$        | $+9$      |

The slope changes sign at 30, so  $f$  is minimised at  $x^* = 30$  (the weighted median: 6 of the 9 trucks go to towns at or left of 30, 6 at or right of it), with

$$f(30) = 2 \cdot 30 + 1 \cdot 10 + 3 \cdot 0 + 1 \cdot 40 + 2 \cdot 70 = 250 \text{ truck-km.}$$

Certificate:  $f$  is convex, decreasing before 30 and increasing after.

(d) Add  $50 \leq x \leq 80$ . On  $[50, 80]$  the slope is  $\geq 3 > 0$ , so  $x^* = 50$  and  $f(50) = 100 + 30 + 60 + 20 + 100 = 310$ . Zoning costs 60 truck-km per day.

(e) The constraints only give *lower* bounds  $e_i \geq |x - a_i|$ . Maximising pushes  $e_i \rightarrow \infty$ , so the LP is unbounded, while the true  $\max_{[0, 100]} f$  is finite ( $f(100) = 2 \cdot 100 + 80 + 3 \cdot 70 + 30 = 520$ ). Minimisation pushes  $e_i$  down onto  $|x - a_i|$ ; maximisation pushes it away. Maximising a convex function is not an LP (Problem 11(c) proves that no LP can do it); it needs integer variables.

### Solution to Problem 2

(a)

$$\min t \quad \text{s.t.} \quad y_i - \alpha - \beta x_i \leq t, \quad -(y_i - \alpha - \beta x_i) \leq t \quad (i = 1, \dots, 4).$$

$\alpha, \beta$  are free (a line may have negative slope or intercept);  $t \geq 0$  is implied. Equational form: split  $\alpha = \alpha^+ - \alpha^-$ ,  $\beta = \beta^+ - \beta^-$  and add 8 slacks:  $5 + 8 = 13$  variables, 8 equations.

(b) With  $\rho_2 = 2 - \alpha - \beta$ ,  $\rho_3 = 1 - \alpha - 2\beta$ ,  $\rho_4 = 4 - \alpha - 3\beta$  (points at  $x = 1, 2, 3$ ), take  $\rho_2 \leq t$ ,  $-\rho_3 \leq t$ ,  $\rho_4 \leq t$  with weights  $\lambda = (1, 2, 1)$ . The weights are found by requiring the  $\alpha$ - and  $\beta$ -coefficients of  $\lambda_1 \rho_2 - \lambda_2 \rho_3 + \lambda_3 \rho_4$  to vanish:  $-\lambda_1 + \lambda_2 - \lambda_3 = 0$  and  $-\lambda_1 + 2\lambda_2 - 3\lambda_3 = 0$ , giving  $\lambda \propto (1, 2, 1)$ . Then

$$\rho_2 - 2\rho_3 + \rho_4 = (2 - \alpha - \beta) - 2(1 - \alpha - 2\beta) + (4 - \alpha - 3\beta) = 4,$$

so  $4 = \rho_2 - 2\rho_3 + \rho_4 \leq t + 2t + t = 4t$ , i.e.  $t \geq 1$  for every line.

(c)  $t = 1$  requires equality in all three inequalities:  $\rho_2 = 1$ ,  $\rho_3 = -1$ ,  $\rho_4 = 1$ , i.e.  $\alpha + \beta = 1$ ,  $\alpha + 2\beta = 2$ ,  $\alpha + 3\beta = 3$ , so  $\beta = 1$ ,  $\alpha = 0$  (the third equation is then satisfied). The remaining point has  $\rho_1 = 0$ ,  $|\rho_1| \leq 1$ . Hence the line  $y = x$  is optimal with maximum error 1, and it is

unique because *any* optimal line must satisfy the three equalities, which have one solution. Note the residuals  $+1, -1, +1$ : the errors alternate in sign (equioscillation).

(d)  $\min \sum_i e_i$  s.t.  $e_i \geq y_i - \alpha - \beta x_i$ ,  $e_i \geq -(y_i - \alpha - \beta x_i)$ . Least squares has a quadratic objective, which is not linear (and has no kink to linearise:  $\rho^2$  is not a maximum of finitely many affine functions).

### Solution to Problem 3

(a)  $C(E)$  has slope 1, 3, 7 on the three blocks, so it is piecewise linear with kinks at 40 and 100. Profit  $= 13x_1 + 9x_2 - C(2x_1 + x_2)$  is therefore not linear.

(b)

$$\begin{aligned} \max \quad & 13x_1 + 9x_2 - e_1 - 3e_2 - 7e_3 \\ \text{s.t.} \quad & e_1 + e_2 + e_3 = 2x_1 + x_2, \quad x_1 + x_2 \leq 60, \quad x_2 \leq 40, \\ & 0 \leq e_1 \leq 40, \quad 0 \leq e_2 \leq 60, \quad e_3 \geq 0, \quad x \geq 0. \end{aligned}$$

(c) Fix  $x$ , hence  $E$ . Among splits  $(e_1, e_2, e_3)$  with sum  $E$  and within the bounds, suppose  $e_2 > 0$  while  $e_1 < 40$ . Moving  $\varepsilon = \min(e_2, 40 - e_1) > 0$  from block 2 to block 1 keeps feasibility and lowers the cost by  $(3 - 1)\varepsilon > 0$ . The same holds for  $e_3 > 0$  with  $e_1 < 40$  (saving  $6\varepsilon$ ) or  $e_2 < 60$  (saving  $4\varepsilon$ ). So the cheapest split fills block 1, then 2, then 3, and its cost is exactly  $C(E)$ . Hence the LP value equals  $\max_x (13x_1 + 9x_2 - C(E))$ , the real problem. The argument works because the block prices are *increasing*, i.e.  $C$  is convex.

(d) Check that  $13x_1 + 9x_2 = 4(2x_1 + x_2) + 5(x_1 + x_2) = 4E + 5L$ . Then

$$\text{profit} = 5L + (4E - C(E)) \leq 5 \cdot 60 + \max_E g(E), \quad g(E) = 4E - C(E).$$

$g$  has slopes  $4 - 1 = 3$ ,  $4 - 3 = 1$ ,  $4 - 7 = -3$ , so it is maximised at  $E = 100$  with  $g(100) = 400 - (40 + 180) = 180$ . Hence profit  $\leq 300 + 180 = 480$ . Equality needs  $L = 60$  and  $E = 100$ :  $x_1 = E - L = 40$ ,  $x_2 = 2L - E = 20 \leq 40$ .

$x_1 = 40 \text{ A-lamps, } x_2 = 20 \text{ B-lamps, } 100 \text{ kWh, profit } 700 - 220 = 480 \text{ DH, unique.}$

Interpretation: switching a B to an A uses 1 more kWh and earns 4 DH more. That is worth it at 3 DH/kWh but not at 7 DH/kWh, so production stops exactly at the 100 kWh kink.

(e) With blocks  $e_1 \in [0, 40]$  at 5 and  $e_2 \geq 0$  at 2, the LP puts all energy in  $e_2$  and pays  $2E$ , whereas the real bill is  $5E$  for  $E \leq 40$  and  $200 + 2(E - 40)$  beyond. The LP underestimates the cost by up to 120 DH and may choose a wrong plan. The exchange argument of (c) now runs the other way (moving energy to the cheaper later block *saves* money). The tariff is concave; minimising a concave cost needs a binary variable  $z$  with  $e_2 \leq Mz$ ,  $e_1 \geq 40z$  (an integer program).

### Solution to Problem 4

(a)  $R$  is a quotient of affine functions, not an affine function. Along  $x_1 = 0$ ,  $R(0, x_2) = \frac{3x_2+1}{x_2+2}$  takes the values  $\frac{1}{2}, \frac{4}{3}, \frac{7}{4}$  at  $x_2 = 0, 1, 2$ : the increments  $\frac{5}{6}$  and  $\frac{5}{12}$  differ, so  $R$  is not affine.

(b) Multiply numerator, constraints and the definition of  $t$  by  $t > 0$ :  $t(2x_1 + 3x_2 + 1) = 2y_1 + 3y_2 + t$ ;  $t(x_1 + 3x_2) \leq 9t$  gives  $y_1 + 3y_2 \leq 9t$ ; similarly  $2y_1 + y_2 \leq 8t$ ; and  $t(x_1 + x_2 + 2) = 1$  gives  $y_1 + y_2 + 2t = 1$ . Nonnegativity carries over.

(c) *Every feasible  $(y, t)$  has  $t > 0$* : if  $t = 0$  then  $y_1 + 3y_2 \leq 0$  with  $y \geq 0$  forces  $y = 0$ , and then  $y_1 + y_2 + 2t = 0 \neq 1$ .

*Forward map  $x \mapsto (y, t) = (x, 1)/(x_1 + x_2 + 2)$* : shown feasible in (b), with objective  $R(x)$ .

*Backward map*  $(y, t) \mapsto x = y/t$  (legal since  $t > 0$ ):  $x \geq 0$ ; dividing the constraints by  $t$  gives  $x_1 + 3x_2 \leq 9$ ,  $2x_1 + x_2 \leq 8$ ; and  $x_1 + x_2 + 2 = (y_1 + y_2 + 2t)/t = 1/t$ , so  $R(x) = t(2x_1 + 3x_2 + 1) = 2y_1 + 3y_2 + t$ .

The maps are inverse to each other and preserve the objective, so optimal values and optimal solutions correspond.

(d) Since

$$R(x) \leq 2 \iff 2x_1 + 3x_2 + 1 \leq 2x_1 + 2x_2 + 4 \iff x_2 \leq 3,$$

and  $x_2 \leq 3$  follows from  $x_1 + 3x_2 \leq 9$ ,  $x_1 \geq 0$ , we get  $R \leq 2$  everywhere.  $R(0, 3) = 10/5 = 2$ , so  $x^* = (0, 3)$  is optimal, and it is unique ( $R = 2$  forces  $x_2 = 3$ , then  $x_1 \leq 0$ ). In the transformed LP:  $t^* = \frac{1}{5}$ ,  $y^* = (0, \frac{3}{5})$ , value  $\frac{9}{5} + \frac{1}{5} = 2$ , certified by

$$2y_1 + 3y_2 + t \leq \frac{1}{3}(y_1 + 3y_2 - 9t) + 2(y_1 + y_2 + 2t) \leq 0 + 2$$

(compare coefficients:  $\frac{7}{3} \geq 2$ ,  $3 \geq 3$ ,  $1 \geq 1$ , and use  $y \geq 0$ ). For comparison, the other vertices give  $R(0, 0) = \frac{1}{2}$ ,  $R(4, 0) = \frac{3}{2}$ ,  $R(3, 2) = \frac{13}{7}$ .

(e) In defining  $t$  and in multiplying inequalities by  $t$ : if the denominator could vanish,  $t$  is undefined; if it could be negative, multiplying would reverse the inequalities.

## Solution to Problem 5

(a) Let  $g, b, m \geq 0$  be kg of each tea, total  $T = g + b + m$ . Profit per kg:  $55 - 60 = -5$ ,  $55 - 40 = 15$ ,  $55 - 25 = 30$ .

$$\begin{aligned} \max \quad & -5g + 15b + 30m \\ \text{s.t.} \quad & g \geq 0.25(g + b + m) \iff -3g + b + m \leq 0, \\ & m \leq 0.4(g + b + m) \iff -2g - 2b + 3m \leq 0, \\ & g \leq 30, \quad b \leq 50, \quad m \leq 80, \quad g, b, m \geq 0. \end{aligned}$$

The rule  $g/T \geq 0.25$  is not linear; multiplying by  $T > 0$  gives an equivalent linear inequality. For  $T = 0$  the ratio is undefined, but the linear version accepts  $(0, 0, 0)$ , which is the harmless plan “produce nothing”. So the linear model is exactly right.

(b) From the first rule,  $g + b + m \leq 4g \leq 4 \cdot 30 = 120$ .

(c) Guess that the green stock and both recipe rules bind:  $g = 30$ ,  $T = 120$ ,  $m = 0.4 \cdot 120 = 48$ ,  $b = 120 - 30 - 48 = 42 \leq 50$ . Profit =  $-150 + 630 + 1440 = 1920$ . Multipliers  $y = (21, 3, 64)$  on (green rule, mint rule, green stock):

$$21(-3g + b + m) + 3(-2g - 2b + 3m) + 64g = -5g + 15b + 30m$$

(check:  $-63 - 6 + 64 = -5$ ,  $21 - 6 = 15$ ,  $21 + 9 = 30$ ). Since the first two brackets are  $\leq 0$  and  $g \leq 30$ ,

$$-5g + 15b + 30m \leq 64 \cdot 30 = 1920$$

for every feasible plan. The guess achieves it, so it is optimal:

30 kg green, 42 kg black, 48 kg mint: 120 kg of blend, profit 1920 DH.

(The multipliers solve  $y_1 - 2y_2 = 15$ ,  $y_1 + 3y_2 = 30$ ,  $-3y_1 - 2y_2 + y_3 = -5$ , i.e. equality on the variables that are positive.)

(d) Each kg of green “unlocks” 3 more kg of other teas through the 25% rule, and those are very profitable. The multiplier  $y_3 = 64$  is the value of one more kg of green in stock: re-solving with 31 kg gives  $(31, 43.4, 49.6)$  and profit  $1984 = 1920 + 64$ . So the shop should pay up to 64 DH above list price per extra kg.

## Solution to Problem 6

(a) Arc flows  $x_{sa}, x_{sb}, x_{ac}, x_{at}, x_{bc}, x_{ct}$ :

$$\begin{aligned} \max \quad & x_{sa} + x_{sb} \\ \text{s.t.} \quad & x_{sa} = x_{ac} + x_{at}, \quad x_{sb} = x_{bc}, \quad x_{ac} + x_{bc} = x_{ct} \quad (\text{conservation at } a, b, c), \\ & x_{ac} + x_{bc} \leq 10 \quad (\text{router}), \\ & 0 \leq x_{sa} \leq 10, \quad 0 \leq x_{sb} \leq 8, \quad 0 \leq x_{ac} \leq 7, \quad 0 \leq x_{at} \leq 4, \quad 0 \leq x_{bc} \leq 9, \quad 0 \leq x_{ct} \leq 15. \end{aligned}$$

(b) In the split network, conservation at  $c_{\text{in}}$  and  $c_{\text{out}}$  reads  $x_{ac} + x_{bc} = x_c = x_{ct}$ , where  $x_c \leq 10$  is the new arc. This is exactly the constraint set of (a) with the new name  $x_c$  for the router throughput. For the LP alone nothing is gained, since (a) is already linear. What the split buys is a network with *only arc capacities*, to which the cut bound (Theorem 1.6.12) and max-flow/min-cut reasoning apply directly: a cut may now “cut the router”.

(c) Flow:  $x_{sa} = 6, x_{sb} = 8, x_{at} = 4, x_{ac} = 2, x_{bc} = 8, x_c = x_{ct} = 10$ . Value 14. Cut  $S = \{s, a, b, c_{\text{in}}\}$ : the arcs leaving  $S$  are  $a \rightarrow t$  (4) and  $c_{\text{in}} \rightarrow c_{\text{out}}$  (10), capacity 14. Since every flow is at most every cut capacity, 14 is the maximum.

(d) The cut  $S$  above does not contain  $s \rightarrow a, s \rightarrow b, a \rightarrow c, b \rightarrow c, c \rightarrow t$ . Raising any of these leaves the capacity of  $S$  at 14, so the maximum flow stays 14. The two arcs in the cut both help:

- $a \rightarrow t$  raised to 5:  $x_{sa} = 7, x_{at} = 5, x_{ac} = 2, x_{sb} = x_{bc} = 8, x_c = x_{ct} = 10$ : value 15.
- router raised to 11:  $x_{sa} = 7, x_{at} = 4, x_{ac} = 3, x_{sb} = x_{bc} = 8, x_c = x_{ct} = 11$ : value 15.

(These are maximum, since the cut now has capacity 15.) Upgrading the input links or the big  $c \rightarrow t$  link is useless.

## Solution to Problem 7

(a)  $R_3 - R_1 - R_2$  gives  $[0 \ 0 \ 0 \ 0 \ 0 \mid 12 - 4 - 8] = [0 \cdots 0 \mid 0]$ : a zero row. So  $\text{rank } A = 2$  (the first two rows are clearly independent) and the third equation is redundant; delete it (Proposition 2.3.4). With right-hand side 13 the row becomes  $[0 \cdots 0 \mid 1]$ , i.e.  $0 = 1$ : the system is inconsistent and the LP infeasible (Theorem 2.3.3).

(b) Columns  $a_1 = (1, 2), a_2 = (1, 1), a_3 = (1, 0), a_4 = (0, 1), a_5 = (2, 2) = 2a_2$ .

| pair       | $\det A_B$ | basic solution $(x_1, \dots, x_5)$ | status                                     |
|------------|------------|------------------------------------|--------------------------------------------|
| $\{1, 2\}$ | -1         | $(4, 0, 0, 0, 0)$                  | feasible, degenerate                       |
| $\{1, 3\}$ | -2         | $(4, 0, 0, 0, 0)$                  | feasible, degenerate                       |
| $\{1, 4\}$ | 1          | $(4, 0, 0, 0, 0)$                  | feasible, degenerate                       |
| $\{1, 5\}$ | -2         | $(4, 0, 0, 0, 0)$                  | feasible, degenerate                       |
| $\{2, 3\}$ | -1         | $(0, 8, -4, 0, 0)$                 | infeasible                                 |
| $\{2, 4\}$ | 1          | $(0, 4, 0, 4, 0)$                  | feasible                                   |
| $\{2, 5\}$ | 0          | —                                  | not a basis ( $x_2 + 2x_5 = 4$ and $= 8$ ) |
| $\{3, 4\}$ | 1          | $(0, 0, 4, 8, 0)$                  | feasible                                   |
| $\{3, 5\}$ | 2          | $(0, 0, -4, 0, 4)$                 | infeasible                                 |
| $\{4, 5\}$ | -2         | $(0, 0, 0, 4, 2)$                  | feasible                                   |

Example computation,  $\{3, 5\}$ :  $x_3 + 2x_5 = 4, 2x_5 = 8$ , so  $x_5 = 4, x_3 = -4$ .

(c)  $10 = 1$  (singular: dependent columns  $a_5 = 2a_2$ ) + 2 (bases with a negative component) + 7 (feasible bases). The 7 feasible bases give only 4 points because  $(4, 0, 0, 0, 0)$  is degenerate: its support  $\{1\}$  has size  $1 < m$ , and it extends to a basis in 4 ways.

(d) Objective values:  $20, 8+4=12, 4+8=12, 4+6=10$ . By the fundamental theorem (feasible, and bounded since the equations give  $x_1 \leq 4, x_2+x_3+2x_5 \leq 4, x_4 \leq 8$ ) the best BFS  $(4, 0, 0, 0, 0)$  is optimal. The four objective rows:

| basis          | objective row                                     |
|----------------|---------------------------------------------------|
| $\{x_1, x_2\}$ | $v = 20 + 2x_3 - 2x_4 - x_5$                      |
| $\{x_1, x_3\}$ | $v = 20 - x_2 - x_4 - 3x_5$                       |
| $\{x_1, x_4\}$ | $v = 20 - 2x_2 - 2x_3 - 5x_5$                     |
| $\{x_1, x_5\}$ | $v = 20 + \frac{1}{2}x_2 + 3x_3 - \frac{5}{2}x_4$ |

(For  $\{x_1, x_4\}$ :  $x_1 = 4 - x_2 - x_3 - 2x_5$  and  $x_4 = x_2 + 2x_3 + 2x_5$ , substitute into the objective.) Only  $\{x_1, x_3\}$  and  $\{x_1, x_4\}$  pass the optimality test.  $\{x_1, x_4\}$  has all  $r_j < 0$ , so  $x^*$  is the *unique* optimum (Theorem 3.3.8). The other two bases show positive reduced costs at an optimal point.

(e) Tableau  $\{x_1, x_2\}$ :  $x_1 = 4 + x_3 - x_4, x_2 = 0 - 2x_3 + x_4 - 2x_5$ .  $x_3$  enters ( $r_3 = 2$ ). The row of  $x_2$  limits  $x_3 \leq 0/2 = 0$  and the row of  $x_1$  does not limit, so  $\theta^* = 0$  and  $x_2$  leaves. The new basis  $\{x_1, x_3\}$  describes the same point with the same value 20, and now all reduced costs are  $\leq 0$ : a zero step, then stop. The positive  $r_3$  in  $\{x_1, x_2\}$  pointed in a direction that is infeasible from the start, because the basic variable  $x_2$  is already 0.

## Solution to Problem 8

(a) Slacks  $s_1, s_2, s_3$ .

$$\begin{array}{lll}
 s_1 = 8 - x_1 - 2x_2 & s_1 = 2 - x_1 + 2s_3 & x_1 = 2 - s_1 + 2s_3 \\
 \text{T0: } s_2 = 4 - x_1 & \text{T1: } s_2 = 4 - x_1 & \text{T2: } s_2 = 2 + s_1 - 2s_3 \\
 s_3 = 3 - x_2 & x_2 = 3 - s_3 & x_2 = 3 - s_3 \\
 v = 0 + 2x_1 + 4x_2 & v = 12 + 2x_1 - 4s_3 & v = 16 - 2s_1 + 0s_3
 \end{array}$$

T0→T1:  $x_2$  enters (4), ratios  $s_1:4, s_3:3$ , so  $s_3$  leaves,  $\theta^* = 3$ . T1→T2:  $x_1$  enters, ratios  $s_1:2, s_2:4$ , so  $s_1$  leaves,  $\theta^* = 2$ . T2 passes the test:  $(x_1, x_2) = (2, 3), v = 16$ .

(b)  $s_3$  has  $r = 0$ . Ratios:  $s_2 = 2 - 2s_3$  gives 1;  $x_2 = 3 - s_3$  gives 3;  $x_1$  increases. So  $s_2$  leaves,  $\theta^* = 1$ :

$$x_1 = 4 - s_2, \quad s_3 = 1 + \frac{1}{2}s_1 - \frac{1}{2}s_2, \quad x_2 = 2 - \frac{1}{2}s_1 + \frac{1}{2}s_2, \quad v = 16 - 2s_1 + 0s_2.$$

The point moves to  $(4, 2)$ , the basis changes, the value stays 16. (Pivoting on  $s_2$  would take you back.)

(c) From T2,  $v = 16 - 2s_1$  holds for *every* feasible point. So  $v \leq 16$ , with equality iff  $s_1 = 0$ , i.e.  $x_1 + 2x_2 = 8$ . Intersecting that line with the other constraints:  $x_1 = 8 - 2x_2 \leq 4$  gives  $x_2 \geq 2$ , and  $x_2 \leq 3$ . Hence

$$X^* = \{(8 - 2\lambda, \lambda) : 2 \leq \lambda \leq 3\},$$

the segment from  $(2, 3)$  to  $(4, 2)$ : two optimal BFS and infinitely many optimal points in between.

(d) Recompute the objective row of T2 with  $c_1x_1 + 4x_2$ :

$$v = c_1(2 - s_1 + 2s_3) + 4(3 - s_3) = (12 + 2c_1) - c_1s_1 + (2c_1 - 4)s_3.$$

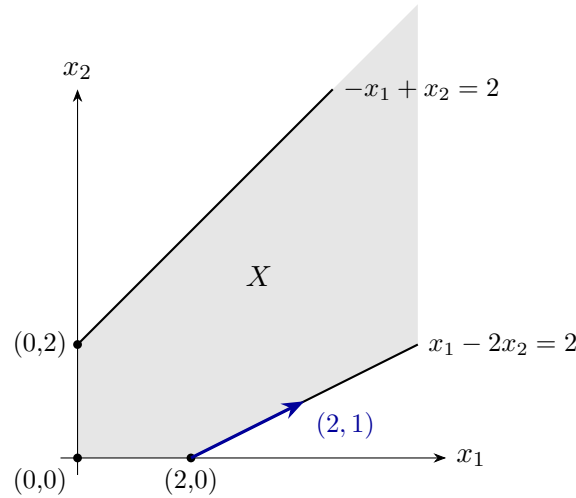
The test holds iff  $c_1 \geq 0$  and  $2c_1 - 4 \leq 0$ :  $(2, 3)$  is optimal exactly for  $0 \leq c_1 \leq 2$  (uniquely for  $0 < c_1 < 2$ ). For  $c_1$  slightly above 2,  $s_3$  has positive reduced cost, and the pivot of (b) leads to  $(4, 2)$ . Recomputing the objective row of that tableau,

$$v = c_1(4 - s_2) + 4(2 - \frac{1}{2}s_1 + \frac{1}{2}s_2) = (4c_1 + 8) - 2s_1 + (2 - c_1)s_2,$$

and for  $c_1 > 2$  both coefficients are negative:  $(4, 2)$  is the unique optimum. Geometrically: the objective rotates past the direction of the edge  $x_1 + 2x_2 = 8$ .

## Solution to Problem 9

(a)



Vertices  $(0,0)$ ,  $(2,0)$ ,  $(0,2)$ . The two slanted lines would meet at  $x_2 = -4$ , outside  $x \geq 0$ , so the region opens up to the upper right: e.g.  $(2 + 2\theta, \theta) \in X$  for all  $\theta \geq 0$  (check:  $-(2 + 2\theta) + \theta = -2 - \theta \leq 2$  and  $2 + 2\theta - 2\theta = 2 \leq 2$ ). Hence  $X$  is unbounded.

(b)  $x + \theta d \in X$  for all  $\theta \geq 0$  and some  $x \in X$  forces  $d \geq 0$ ,  $-d_1 + d_2 \leq 0$ ,  $d_1 - 2d_2 \leq 0$  (otherwise a constraint is violated for large  $\theta$ ); conversely these conditions keep every constraint satisfied along the ray. They say  $d_2 \leq d_1 \leq 2d_2$ . Put  $\lambda = 2d_2 - d_1 \geq 0$ ,  $\mu = d_1 - d_2 \geq 0$ ; then  $\lambda(1,1) + \mu(2,1) = (d_1, d_2)$ . Conversely every such combination satisfies the inequalities.

(c) Equational form:  $-x_1 + x_2 + s_1 = 2$ ,  $x_1 - 2x_2 + s_2 = 2$ . At  $(2,0)$ :  $x_1 = 2$ ,  $s_1 = 4$ ,  $x_2 = s_2 = 0$ . From row 2,  $x_1 = 2 + 2x_2 - s_2$ ; then  $s_1 = 2 + x_1 - x_2 = 4 + x_2 - s_2$ :

$$x_1 = 2 + 2x_2 - s_2, \quad s_1 = 4 + x_2 - s_2, \quad v = x_1 + \alpha x_2 = 2 + (2 + \alpha)x_2 - s_2.$$

- If  $\alpha > -2$ :  $r_{x_2} = 2 + \alpha > 0$  and the column of  $x_2$  is  $(+2, +1) \geq 0$ , so by Theorem 3.3.9 the LP is unbounded along  $x(\theta) = (2 + 2\theta, \theta)$ ,  $v = 2 + (2 + \alpha)\theta \rightarrow \infty$ . The direction is  $(2, 1)$ , the recession direction with  $c^\top d = 2 + \alpha > 0$ .
- If  $\alpha \leq -2$ : all reduced costs are  $\leq 0$ , so  $(2, 0)$  is optimal with value 2.

(d) Directly,  $x_1 + \alpha x_2 = (x_1 - 2x_2) + (\alpha + 2)x_2 \leq 2 + 0$  when  $\alpha \leq -2$ .

- $\alpha < -2$ : equality needs  $x_1 - 2x_2 = 2$  and  $x_2 = 0$ , so  $X^* = \{(2, 0)\}$  (also: both  $r < 0$ ).
- $\alpha = -2$ : equality iff  $x_1 - 2x_2 = 2$ , so  $X^* = \{(2 + 2\theta, \theta) : \theta \geq 0\}$ , a whole ray.

(e) No contradiction. The fundamental theorem says that if an optimum exists, *some* BFS is optimal ( $(2, 0)$  here); it does not say optimal solutions are bounded or unique. In the tableau,  $r_{x_2} = 0$  with a nonnegative column: moving along  $x_2$  never hits a constraint and never changes  $v$  (Problem 15(c)). It is case 1 of Theorem 3.3.9 with  $r_j = 0$  instead of  $r_j > 0$ .

## Solution to Problem 10

(a)  $x_1 + x_2 - s_1 = 4$ ,  $x_1 + 3x_2 - s_2 = 6$ ,  $x_1 + s_3 = 5$ , all variables  $\geq 0$ . The basis  $\{s_1, s_2, s_3\}$  gives  $s_1 = -4$ ,  $s_2 = -6$ ,  $s_3 = 5$ : infeasible.

(b) Rows:  $x_1 + x_2 - s_1 + a_1 = 4$ ,  $x_1 + 3x_2 - s_2 + a_2 = 6$ ,  $x_1 + s_3 = 5$ . Since  $b \geq 0$ , the basis  $\{a_1, a_2, s_3\}$  gives  $(a_1, a_2, s_3) = (4, 6, 5) \geq 0$ : feasible.  $w = -a_1 - a_2 \leq 0$  always, so the optimum is  $\leq 0$  (and bounded). If the original LP has a feasible  $x$ , then  $(x, s, a = 0)$  is feasible for the

auxiliary LP with  $w = 0$ , so the optimum is 0. Conversely, an auxiliary solution with  $w = 0$  has  $a = 0$ , and its  $(x, s)$  satisfies the original equations.

(c) Phase I tableaux:

$$\begin{aligned} \text{T0: } a_1 &= 4 - x_1 - x_2 + s_1, & a_2 &= 6 - x_1 - 3x_2 + s_2, & s_3 &= 5 - x_1, \\ w &= -10 + 2x_1 + 4x_2 - s_1 - s_2. \end{aligned}$$

$x_2$  enters; ratios  $a_1:4, a_2:2$ ;  $a_2$  leaves ( $\theta^* = 2$ ), then drop  $a_2$ :

$$\begin{aligned} \text{T1: } a_1 &= 2 - \frac{2}{3}x_1 + s_1 - \frac{1}{3}s_2, & x_2 &= 2 - \frac{1}{3}x_1 + \frac{1}{3}s_2, & s_3 &= 5 - x_1, \\ w &= -2 + \frac{2}{3}x_1 - s_1 + \frac{1}{3}s_2. \end{aligned}$$

$x_1$  enters; ratios  $a_1:3, x_2:6, s_3:5$ ;  $a_1$  leaves ( $\theta^* = 3$ ), then drop  $a_1$ :

$$\text{T2: } x_1 = 3 + \frac{3}{2}s_1 - \frac{1}{2}s_2, \quad x_2 = 1 - \frac{1}{2}s_1 + \frac{1}{2}s_2, \quad s_3 = 2 - \frac{3}{2}s_1 + \frac{1}{2}s_2, \quad w = 0.$$

$w^* = 0$ : the original LP is feasible and  $\{x_1, x_2, s_3\}$  is a feasible basis, BFS  $(3, 1)$ .

(d) Maximise  $v = -3x_1 - 2x_2$ . Substituting T2:

$$v = -3(3 + \frac{3}{2}s_1 - \frac{1}{2}s_2) - 2(1 - \frac{1}{2}s_1 + \frac{1}{2}s_2) = -11 - \frac{7}{2}s_1 + \frac{1}{2}s_2.$$

$s_2$  enters; only the row of  $x_1$  limits it ( $3 - \frac{1}{2}s_2 \geq 0$ , ratio 6);  $x_1$  leaves:

$$s_2 = 6 - 2x_1 + 3s_1, \quad x_2 = 4 - x_1 + s_1, \quad s_3 = 5 - x_1, \quad v = -8 - x_1 - 2s_1.$$

All reduced costs are  $< 0$ : the unique optimum is  $x^* = (0, 4)$  with cost 8. Certificate:  $3x_1 + 2x_2 = 2(x_1 + x_2) + x_1 \geq 2 \cdot 4 + 0 = 8$ . (Phase I stopped at  $(3, 1)$ , cost 11; Phase II moved to  $(0, 4)$ .)

(e) Now the third row is  $x_1 + x_2 + s_3 = 3$ . T0 is as before except  $s_3 = 3 - x_1 - x_2$ ;  $x_2$  enters and  $a_2$  leaves (ratios 4, 2, 3). Then

$$a_1 = 2 - \frac{2}{3}x_1 + s_1 - \frac{1}{3}s_2, \quad x_2 = 2 - \frac{1}{3}x_1 + \frac{1}{3}s_2, \quad s_3 = 1 - \frac{2}{3}x_1 - \frac{1}{3}s_2, \quad w = -2 + \frac{2}{3}x_1 - s_1 + \frac{1}{3}s_2.$$

$x_1$  enters; ratios  $a_1:3, x_2:6, s_3:\frac{3}{2}$ ;  $s_3$  leaves:

$$x_1 = \frac{3}{2} - \frac{1}{2}s_2 - \frac{3}{2}s_3, \quad x_2 = \frac{3}{2} + \frac{1}{2}s_2 + \frac{1}{2}s_3, \quad a_1 = 1 + s_1 + s_3, \quad w = -1 - s_1 + 0s_2 - s_3.$$

Optimal with  $w^* = -1 < 0$ : the LP is *infeasible*. The last row even says  $a_1 = 1 + s_1 + s_3 \geq 1$  for every solution of the equations, which is the contradiction between  $x_1 + x_2 \geq 4$  and  $x_1 + x_2 \leq 3$ .

## Solution to Problem 11

(a) Write  $L$  for the left infimum and  $R$  for the right one. If  $(x, e)$  is feasible on the right, then  $e_i \geq \max(f_i(x), -f_i(x)) = |f_i(x)|$ , so  $\sum e_i \geq \sum |f_i(x)| \geq L$ ; hence  $R \geq L$ . For every  $x \in X$ ,  $(x, e)$  with  $e_i = |f_i(x)|$  is feasible with value  $\sum |f_i(x)|$ ; hence  $R \leq L$ . So  $L = R$ .

If  $(x, e)$  is optimal on the right:  $L = R = \sum e_i \geq \sum |f_i(x)| \geq L$ , so both are equalities, which gives  $x$  optimal on the left and  $\sum(e_i - |f_i(x)|) = 0$  with every term  $\geq 0$ , i.e.  $e_i = |f_i(x)|$ . Conversely, if  $x$  is optimal on the left,  $(x, |f(x)|)$  has value  $L = R$ , so it is optimal on the right.

(b) *Min-max*:  $\min_{x \in X} \max_i f_i(x) = \inf\{t : x \in X, t \geq f_i(x) \forall i\}$ , with  $(x, t)$  optimal iff  $x$  optimal and  $t = \max_i f_i(x)$ . Proof:  $t \geq f_i(x) \forall i \iff t \geq \max_i f_i(x)$ ; then repeat (a) with one auxiliary variable. *Max-min*:  $\max_x \min_i f_i(x) = \sup\{t : t \leq f_i(x) \forall i\}$ , same proof with inequalities reversed. The pattern: the auxiliary variable must be pushed by the objective *towards* the kinked function, i.e. minimise a convex function or maximise a concave one.

(c)  $\max e$  s.t.  $e \geq x, e \geq -x, -1 \leq x \leq 1$ : the point  $(0, e)$  is feasible for every  $e \geq 0$ , so the LP is unbounded, while  $\max_{[-1, 1]} |x| = 1$ .

Stronger statement. Suppose an LP in  $(x, z)$  had optimal set  $Y^*$  whose projection  $\{x : (x, z) \in Y^* \text{ for some } z\}$  equals  $\{-1, 1\}$ . By Problem 14(a),  $Y^*$  is convex, and the projection of a convex set is convex: if  $(x, z), (x', z') \in Y^*$  then  $\lambda(x, z) + (1 - \lambda)(x', z') \in Y^*$  projects to  $\lambda x + (1 - \lambda)x'$ . Taking  $x = -1, x' = 1, \lambda = \frac{1}{2}$  puts 0 in the projection, a contradiction.

**Solution to Problem 12**

(a) Using  $x \geq 0$  with  $A^\top y \geq c$ , then  $y \geq 0$  with  $Ax \leq b$ :

$$c^\top x \leq (A^\top y)^\top x = y^\top (Ax) \leq y^\top b.$$

(b) For every feasible  $x'$ :  $c^\top x' \leq b^\top y = c^\top x$ , so  $x$  is optimal. For every certificate  $y'$ :  $b^\top y' \geq c^\top x = b^\top y$ , so  $y$  is minimal.

(c) Expand:

$$b^\top y - c^\top x = y^\top (b - Ax) + x^\top (A^\top y - c) = \sum_i y_i (b - Ax)_i + \sum_j x_j (A^\top y - c)_j.$$

Every term is a product of two nonnegative numbers. If the left side is 0, every term is 0.

(d)  $A^\top y = (2 + 2, 2 + 1, 4 + 1) = (4, 3, 5) = c$  and  $y \geq 0$ : a certificate with  $b^\top y = 20 + 12 = 32$ . In fact  $4x_1 + 3x_2 + 5x_3 = 2(x_1 + x_2 + 2x_3) + (2x_1 + x_2 + x_3) \leq 32$  identically. The point  $(2, 8, 0)$  is feasible ( $10 \leq 10, 12 \leq 12, 2 \leq 9$ ) with value  $8 + 24 = 32$ , so  $v^* = 32$ .

All optimal solutions: by (c), with  $y_1, y_2 > 0$ , an optimal  $x$  has both first constraints tight; conversely any feasible  $x$  with them tight has value 32. Solve  $x_1 + x_2 + 2x_3 = 10, 2x_1 + x_2 + x_3 = 12$ : subtracting,  $x_1 = 2 + x_3$ , then  $x_2 = 8 - 3x_3$ . Put  $x_3 = \lambda$ : need  $\lambda \geq 0, 8 - 3\lambda \geq 0$ , and  $x_1 + 3x_3 = 2 + 4\lambda \leq 9$ , i.e.  $\lambda \leq \frac{7}{4}$  (tighter than  $\frac{8}{3}$ ):

$$X^* = \{(2 + \lambda, 8 - 3\lambda, \lambda) : 0 \leq \lambda \leq \frac{7}{4}\},$$

the segment from  $(2, 8, 0)$  to  $(\frac{15}{4}, \frac{11}{4}, \frac{7}{4})$ .

(e) If a certificate  $y$  existed, (a) would give  $c^\top x \leq b^\top y$  for every feasible  $x$ , a finite upper bound, so  $(P)$  would not be unbounded.

**Solution to Problem 13**

(a) Let  $x$  be the BFS of basis  $B$  ( $x_N = 0, x_B = A_B^{-1}b$ ) and suppose  $x = \lambda u + (1 - \lambda)w$  with  $u, w \in P, 0 < \lambda < 1$ . For  $j \in N$ :  $0 = \lambda u_j + (1 - \lambda)w_j$  with both terms  $\geq 0$ , so  $u_j = w_j = 0$ . Then  $A_B u_B = Au = b$  and  $A_B w_B = b$ , and  $A_B$  is invertible, so  $u_B = w_B = A_B^{-1}b$ . Hence  $u = w = x$ , contradicting  $u \neq w$ .

(b) Let  $S = S(x)$ , and let  $x' \in P$  with  $S(x') \subseteq S$ . Both vanish outside  $S$ , so  $A_S x_S = b = A_S x'_S$ , i.e.  $A_S(x_S - x'_S) = 0$ . The columns of  $A_S$  are independent, so  $x_S = x'_S$  and  $x' = x$ . A BFS has independent support columns (they are among the columns of  $A_B$ ), so two BFS with the same support  $S$  each have support  $\subseteq S$  of the other, hence are equal.

(c) Every BFS is the basic solution of at least one basis, and a basis determines its basic solution ( $x_B = A_B^{-1}b$ ). So  $\#\text{BFS} \leq \#\text{bases} \leq \binom{n}{m}$ . By Theorem 2.4.10, vertices are exactly the BFS.

(d) If the BFS  $x$  of a feasible basis  $B$  is nondegenerate, all  $m$  basic variables are positive and the nonbasic ones are 0, so  $S(x) = B$ . If two feasible bases  $B, B'$  give the same  $x$ , then  $B = S(x) = B'$ . Hence distinct feasible bases give distinct BFS. Without nondegeneracy: in Problem 7, the four bases  $\{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}$  all give  $(4, 0, 0, 0, 0)$ .

**Solution to Problem 14**

(a) Let  $v^*$  be the optimal value and  $x, x' \in X^*, \lambda \in [0, 1]$ .  $X$  is convex:  $A(\lambda x + (1 - \lambda)x') = \lambda Ax + (1 - \lambda)Ax' \leq \lambda b + (1 - \lambda)b = b$ . And  $c^\top(\lambda x + (1 - \lambda)x') = \lambda v^* + (1 - \lambda)v^* = v^*$ . So the combination lies in  $X^*$ .

(b) If there is no optimum, we are done. If  $X^*$  contains two points  $x \neq x'$ , it contains the whole segment  $\{\lambda x + (1 - \lambda)x'\}$ , whose points are distinct for distinct  $\lambda$ : infinitely many.

(c) Suppose some  $\hat{x} \in X$  has  $c^\top \hat{x} > c^\top \bar{x}$ . For  $\lambda \in (0, 1]$ ,  $x_\lambda = \bar{x} + \lambda(\hat{x} - \bar{x}) \in X$  by convexity, and  $c^\top x_\lambda = c^\top \bar{x} + \lambda(c^\top \hat{x} - c^\top \bar{x}) > c^\top \bar{x}$ . Choosing  $\lambda < \varepsilon / \|\hat{x} - \bar{x}\|$  gives  $\|x_\lambda - \bar{x}\| < \varepsilon$ , contradicting local optimality.

(d)  $\max x^2$  on  $[-1, 2]$ :  $x = -1$  is a local maximum (value 1) but the global one is  $x = 2$  (value 4). The step  $c^\top x_\lambda > c^\top \bar{x}$  fails:  $x^2$  is not linear (not concave), so moving towards a better point need not improve. For  $\max x$  over  $X = \{0, 2\}$  (or any integer set), 0 is a local maximum with  $\varepsilon = 1$  but not global: the set is not convex, so  $x_\lambda$  need not be feasible.

## Solution to Problem 15

(a) ( $\Leftarrow$ ) Theorem 3.3.8: if all  $r_j < 0$ ,  $v = v_0$  forces  $x_N = 0$ , hence  $x = \hat{x}$ .

( $\Rightarrow$ ) Contrapositive: suppose  $r_k = 0$  for some  $k \in N$  (the  $r_j$  are  $\leq 0$ , so “not all  $< 0$ ” means some  $= 0$ ). Raise  $x_k = \theta$ , keeping other nonbasics at 0:  $x_i = p_i + q_{ik}\theta$ . Let  $\theta_0 = \min\{p_i / (-q_{ik}) : q_{ik} < 0\}$ , or  $\theta_0 = 1$  if no  $q_{ik} < 0$ . Since all  $p_i > 0$ ,  $\theta_0 > 0$ . For  $0 < \theta \leq \theta_0$  the point is feasible, with  $v = v_0 + 0 \cdot \theta = v_0$ , and differs from  $\hat{x}$  because  $x_k = \theta > 0$ . So the optimum is not unique.

(b) Equational form:  $x_1 + s_1 = 1$ ,  $x_1 + x_2 + s_2 = 1$ . Take the basis  $\{x_1, s_2\}$ :

$$x_1 = 1 - s_1, \quad s_2 = 0 + s_1 - x_2, \quad v = 1 - s_1 + 0x_2.$$

All  $r \leq 0$ , with  $r_{x_2} = 0$ . Yet the optimum is unique:  $v = 1$  forces  $x_1 = 1$ , then  $x_1 + x_2 \leq 1$  forces  $x_2 = 0$ . The proof of (a) fails because  $p_{s_2} = 0$ : raising  $x_2$  is blocked immediately ( $\theta_0 = 0$ ). The BFS  $(1, 0)$  is degenerate.

(c) If  $q_{ik} \geq 0$  for all  $i$ , then  $x_i = p_i + q_{ik}\theta \geq 0$  for all  $\theta \geq 0$ , so the whole ray  $x(\theta)$  is feasible, with constant value  $v_0$ : all its points are optimal. They are distinct and  $x_k = \theta \rightarrow \infty$ , so  $X^*$  is unbounded (as in Problem 9 with  $\alpha = -2$ ).

(d) At a degenerate optimal vertex, the signs of the reduced costs depend on which basis you use:

- all  $r_j \leq 0$  in *some* basis still proves optimality (sufficient condition);
- a positive  $r_j$  does *not* prove non-optimality (Problem 7: bases  $\{x_1, x_2\}$  and  $\{x_1, x_5\}$ );
- a zero  $r_j$  does *not* prove non-uniqueness (part (b));
- all  $r_j < 0$  in *some* basis still proves uniqueness (basis  $\{x_1, x_4\}$  in Problem 7).

In the nondegenerate case, (a) makes the test sharp in both directions.

## Cheat sheet: the tricks

| Situation                 | Linear model                                          | Valid when                        |
|---------------------------|-------------------------------------------------------|-----------------------------------|
| $\min \sum  f_i(x) $      | $\min \sum e_i, e_i \geq \pm f_i(x)$                  | minimising (P1, P11)              |
| $\min \max_i f_i(x)$      | $\min t, t \geq f_i(x)$                               | minimising a max (P2)             |
| $\max \min_i f_i(x)$      | $\max t, t \leq f_i(x)$                               | maximising a min                  |
| convex piecewise cost     | block variables, increasing prices                    | costs convex (P3)                 |
| ratio objective           | $t = 1/\text{den}, y = tx$                            | denominator $> 0$ (P4)            |
| “at least $p\%$ of total” | $x_k \geq p \sum_j x_j$                               | always (P5)                       |
| node capacity             | split node $v_{\text{in}} \rightarrow v_{\text{out}}$ | cut arguments (P6)                |
| lower bound $x \geq \ell$ | $x = \ell + u, u \geq 0$                              | always                            |
| free variable             | $x = x^+ - x^-$                                       | always                            |
| no feasible start         | Phase I with artificials                              | $b \geq 0$ after sign flips (P10) |